

Where Hallucinations Live: A Cross-Architecture Circuit in VQ-Tokenized Vision-Language Models

Shamanthak Hegde^{1*} Xiangrui Liu¹ Maitreya Patel^{2†} Yezhou Yang¹

¹Arizona State University ²Adobe

<https://shamanthak-hegde.github.io/where-hallucinations-live>

Abstract

Unified vision-language models (VLMs) that tokenize images through a vector-quantized (VQ) codebook routinely hallucinate objects on grounded yes/no benchmarks, yet existing decoding-time fixes treat this as generic mis-calibration without an architectural account. Using activation patching across twenty-five models spanning eight LLM families, we identify an early-layer (L_0) attention routing circuit shared across VQ-tokenized VLMs and propose a three-gate diagnostic that distinguishes the models carrying it from those that do not. The diagnostic isolates ten positive models (five natural unified-VQ VLMs across three LLM families and five induced variants) and rejects the remaining fifteen. A single-variable architectural swap (LLaVA-1.6 CLIP+MLP $\rightarrow$ VQ+Linear) installs the circuit, while a matched-compute MLP control on identical data does not, isolating vector quantization as the source of the pathological signal; the routing pathway that carries it is one that the backbone already provides. Against tuned VCD and DoLA baselines, tuned DoLA wins on binary calibration, but **only L_0 ablation reduces object hallucination in open-ended generation** (CHAIR_i reduces by 31 % relatively, whereas tuned DoLA and VCD leave it unchanged or worsen it). These results recast object hallucination in unified VQ VLMs as a property of architecture and pretraining, and yield a targeted intervention that mechanism-agnostic decoding cannot replicate.

1 Introduction

Unified vision-language models that share a single transformer between image and text tokens have become a dominant architecture (Wu et al., 2024; Team, 2024; Wu et al., 2025; Chern et al., 2024; Liu et al., 2024a; Chen et al., 2025; Xie et al., 2024; Wang et al., 2024; Jin et al., 2024). Most of them

tokenize images through a vector-quantized (VQ) codebook so visual tokens occupy the same vocabulary as text. While elegant, this design suffers from a persistent failure mode on grounded yes/no benchmarks such as POPE (Li et al., 2023) and AMBER (Wang et al., 2023): VQ-tokenized models exhibit extreme and architecturally consistent biases. Several, including VILA-U (Wu et al., 2024) and Liquid (Wu et al., 2025), answer "YES" on roughly 78 – 81% of records regardless of image content, while others collapse to the opposite extreme: Chameleon (Team, 2024) answers "YES" on only 22% of records, and Lumina-mGPT (Liu et al., 2024a) on 0.07%. Continuous-projector VLMs such as LLaVA-1.6 (Liu et al., 2024b) and Qwen2.5-VL (Bai et al., 2025) show none of these extreme patterns. The signature is too consistent within the VQ family to be coincidental, and too at odds with the continuous family to be a generic VLM defect, pointing to a shared architectural cause.

Existing interventions for VLM hallucination such as VCD (Leng et al., 2024), DoLA (Chuang et al., 2024), ITI (Li et al., 2024b), VTI (Liu et al., 2024c), operate on output logits without an architectural account. These methods are mechanism-agnostic: each targets a downstream symptom (per-token calibration, layer-contrast logits, residual-direction steering) without explaining *why* a particular family of models hallucinates in the first place. They are also evaluated almost exclusively on binary benchmarks, where a single-token forced-choice makes logit-level fixes maximally effective; whether they actually reduce object hallucination in open-ended generation is a separate question. Recent mechanistic studies of individual VLMs have begun to address the *why* (Olsson et al., 2022; Meng et al., 2022; Wang et al., 2022; Liu et al., 2025), but typically as single-model case studies. Whether a *shared* circuit, a specific attention pathway through which information flows to causally

*Corresponding author: shamanthak@asu.edu

†Partial work done while at Arizona State University.

produce a behavior, underlies hallucination *across* VQ-tokenized architectures, whether such a circuit can be detected before training a model, and whether targeting it yields interventions that beat mechanism-agnostic baselines on generative metrics remain unanswered.

We answer these questions affirmatively. Using activation patching (Meng et al., 2022) across twenty-five models spanning eight LLM families, we identify an early-layer (L_0) attention routing circuit present across five unified-VQ VLMs spanning three LLM families. The circuit is absent in continuous-projector models and is discriminable from other unified-VQ VLMs that fail the diagnostic for mechanistically distinct reasons. It can be constructed from a healthy backbone by a single-variable swap, replacing LLaVA-1.6’s CLIP+MLP projector with a VQ+Linear projector; a matched-compute MLP variant on identical data and budget fails to install it, isolating the VQ bottleneck as the cause.

To formalize the cross-architecture pattern, we propose a three-gate diagnostic combining residual-stream divergence, attention-routing concentration at L_0 , and codebook off-manifold rate, with thresholds tied to architectural properties measured before behavioral outcomes (Section 3.1). The mechanism also yields a targeted intervention. Sweeping VCD and DoLA over 36 configurations on VILA-U reveals a striking dissociation: tuned DoLA wins on binary POPE/AMBER, but **only L_0 ablation reduces object hallucination in open-ended generation** (CHAIR_i 10.3 % $\rightarrow$ 7.1 %). Logit-level fixes calibrate the yes/no decision but do not disrupt the L_0 circuit that injects hallucinated objects during generation. Our main contributions are summarized as follows:

- A cross-architecture L_0 routing circuit present in five natural unified-VQ VLMs across three LLM families, absent in continuous-projector models, and structurally validated on a held-out model not used to calibrate the diagnostic.
- Causal identification of the VQ bottleneck via single-variable construction with a matched-compute MLP control, and a training-time asymmetry showing the circuit is encoded in LLM backbone weights.
- A three-gate diagnostic chain with an architecturally motivated (not threshold-fit) two-tier criterion that correctly discriminates twenty-five models with mechanistically distinct positive and neg-

ative signatures.

- A mechanism-targeted intervention that loses to tuned VCD/DoLA on binary calibration but uniquely reduces object hallucination in open-ended generation.

2 Background and Related Work

Vector quantization in VLMs. VQ tokenizers (Van Den Oord et al., 2017; Esser et al., 2020; Lee et al., 2022; Mentzer et al., 2024; Yu et al., 2024) map each patch of a continuous visual encoder’s output to its nearest entry in a learned codebook of size K , so visual content becomes discrete tokens sharing a vocabulary with text. Two failure modes recur: codebook collapse, where only a small fraction of the K codes are used in practice (Section 4.2); and snapping artifacts, where a small input perturbation flips the output to an unrelated code with no smooth gradient between them, a property that becomes load-bearing for the hallucination mechanism we develop.

Hallucination correction in VLMs. Decoding-time methods correct hallucination through logit-level interventions at inference, without modifying weights: VCD (Leng et al., 2024), DoLA (Chuang et al., 2024), ITI (Li et al., 2024b), and VTI (Liu et al., 2024c). All are architecture-agnostic and offer no account of *why* a given model hallucinates. A separate line of mechanistic-interpretability work addresses the *why*, but as single-model case studies. We extend mechanistic analysis to cross-architecture generality and compare against fully tuned versions of these baselines.

Activation patching and routing circuits. Activation patching (Meng et al., 2022; Wang et al., 2022; Olsson et al., 2022) localizes where behavior is causally encoded: one runs the model on a clean input and a corrupted variant, then re-runs the corrupted input while patching in clean activations one site at a time. Our diagnostic (Section 3.1) builds on this procedure.

3 Method

With reference to Figure 1, our method has three components, each motivated by a question the preceding component leaves open: (i) a mechanistic diagnostic that identifies whether a given VLM has the candidate routing circuit; (ii) a constructive induction procedure with a matched-compute control that tests whether the circuit is causally tied to the

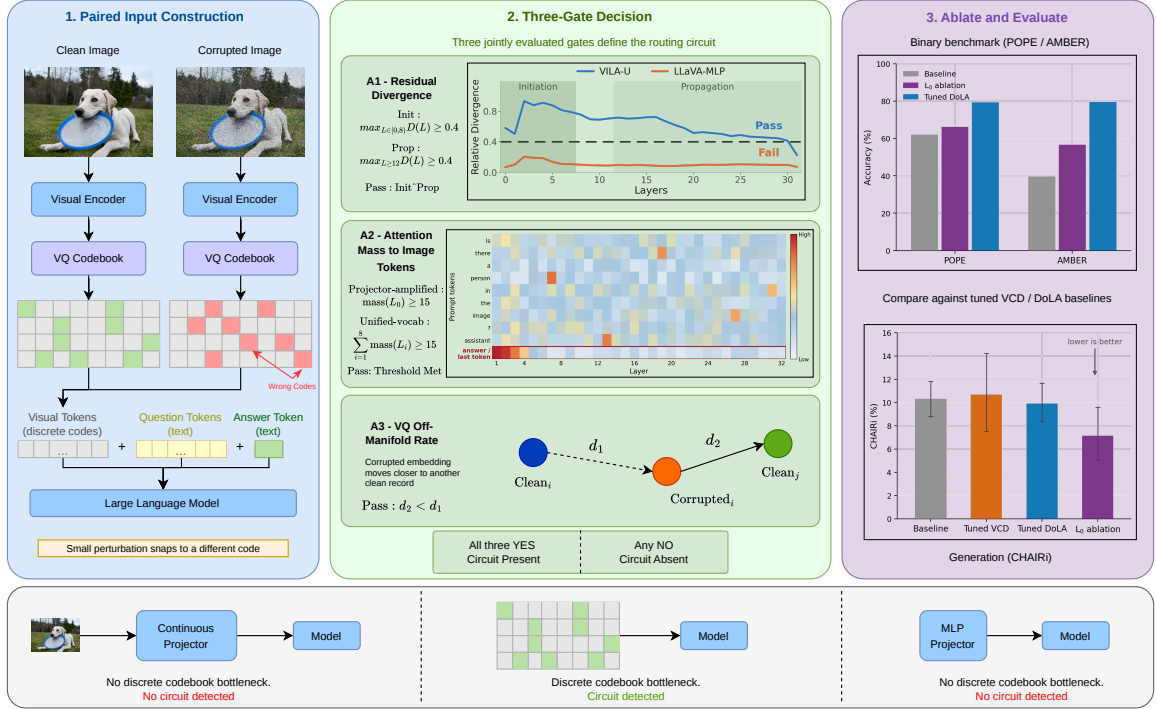


Figure 1: Methodology overview. Clean and corrupted image pairs are constructed and passed through the transformer; residual divergence and early-layer attention mass are measured to detect a routing circuit via a three-gate diagnostic. Models passing all three gates have a VQ-installed L_0 circuit confirmed by a single-variable causal control: swapping only the projector type from MLP to VQ installs the circuit; a matched-compute MLP control does not. Ablating L_0 reduces object hallucination in open-ended generation (CHAIR_i drops 31% relative) while tuned DoLA achieves better binary calibration.

VQ bottleneck, rather than to the incidental features of the models that happen to exhibit it; and (iii) a sanity criterion that distinguishes mechanism-targeted reductions in hallucination from numerical improvements driven by the model failing to respond at all.

3.1 Mechanistic diagnostic: a three-gate chain

We instantiate the activation-patching procedure of Section 2 with two corruption types: Gaussian noise on visual embeddings and image-swap corruption, applied across a fixed set of 500 paired records (300 NaturalBench (Li et al., 2024a) image-swap pairs and 200 POPE (Li et al., 2023) Gaussian-noise pairs). For each record, we run the model on the clean visual embedding v and, depending on corruption type, either a noise version $v + \sigma \cdot \epsilon$ where $\epsilon \sim \mathcal{N}(0, I)$ or a swapped-image embedding, caching activations at every layer; we then re-run the corrupted input while patching in the clean activation at one (token-category, layer) site at a time, with categories visual, question, and answer grouping image-embedding, prompt, and response positions. The restoration score $R =$

$(\text{logit}_{\text{patch}} - \text{logit}_{\text{corrupt}}) / (\text{logit}_{\text{clean}} - \text{logit}_{\text{corrupt}})$ is read out on the answer token: $R = 0, 1, > 1$ correspond to no effect, full restoration, and overshoot.

Per-model noise calibration. Visual-embedding norms vary by over an order of magnitude across our cohort, so a fixed σ would over-perturb some models and under-perturb others. For each model, we sweep σ over a logarithmic grid and select the value whose mean per-token cosine similarity between clean and noisy embeddings, averaged over a subset of POPE probe records, is closest to 0.5; this operates purely on input embeddings, independent of model predictions and labels. Accordingly, σ sorts our cohort into three architectural classes (σ -tiers): *projector-amplified VQ* (high-norm; e.g., VILA-U, Liquid), *continuous-projector* (e.g., LLaVA-1.6, Qwen2.5-VL), and *unified-vocab VQ* (low-norm; e.g., the Chameleon family, where visual tokens enter the LM’s token-embedding table at text-token scale).

Robustness to corruption type. The early-layer divergence that A1 measures is not specific to Gaussian embedding noise: token dropout reproduces

the early peak in both tiers, while permuting the visual-token embeddings produces near-zero divergence despite carrying the largest perturbation magnitude of the corruptions tested, indicating that the routing pools visual-token content independently of order (Appendix A.8).

The three gates. A model passes the diagnostic if it satisfies three gates jointly. **Gate A1: early routing event with downstream propagation.** We measure the per-layer relative divergence between clean and corrupted residual streams (ℓ_2 distance between hidden states, normalized by the clean hidden-state norm). The candidate circuit predicts two things: a routing event concentrated in the early layers, and persistence of the resulting signal rather than its dissipation. A1 therefore has two parts. A1-initiation requires the relative divergence to reach ≥ 0.4 within the early layers $[0, 8)$, the first quarter of the stack. A1-propagation requires it to reach ≥ 0.4 somewhere in the layers $\ell \geq 12$, so that the signal is still carried in the second half of the network. The two measurements are taken on disjoint ranges separated by a four-layer buffer, so neither can be satisfied by the decaying shoulder of the other; the gate tests sustained divergence rather than two separate events, and in several passing models, the elevated divergence is a single continuous ramp or plateau whose onset and persistence the two measurements probe. A model passes A1 only if both hold. A perturbation that produces no early event (initiation fails) reflects a model that does not route visual information through its early layers; one that rises early and then decays below threshold (propagation fails) reflects a routing event that is causally inert. Neither bound is tuned to the cohort partition: the verdict partition is invariant across initiation windows $[0, 4)$ and $[0, 8)$, while at $\ell \geq 8$ the propagation maximum for two positive models falls on the decaying shoulder of the initiation peak, and at $\ell \geq 16$ a positive model drops below threshold. Section 4.3 shows that both failure modes are populated by mechanistically distinct negatives. **Gate A2: early-layer attention concentration.** We measure the fraction of attention mass at layer 0 that the prompt-last token directs to visual tokens, summed over heads. Because the three σ -tiers operate at different embedding-norm regimes, A2 applies a two-tier criterion: L_0 mass ≥ 15 for projector-amplified models; window total over $[0, 8)$ of ≥ 15 for unified-vocab models. The two-tier structure is inherited from the σ -tier clas-

sification (defined from embedding norms before any behavioral outcome is measured) rather than fit to cohort labels. Continuous-projector and decoupled models, which have no early-routing pathway, are measured at the layer-0 sink like projector-amplified models and fall well below the threshold.

Gate A3: codebook off-manifold rate. For each record, we check whether the noised visual embedding is closer to a different record’s nearest codebook entry than to its own. A3 requires this rate to be $\geq 80\%$, capturing the snapping artifact at the cohort scale. A model that cannot be measured at a gate does not pass it, except where the gate is explicitly waived. Where the quantizer exists but uses a fixed lattice, as in the FSQ variants, A3 is waived; where the understanding pathway is not quantized at all, the premise the gate tests is absent, and the model is rejected at this gate.

Why a chain. The gates are applied as a conjunctive chain because, individually, they admit confounds. A2 alone can be passed by a model that routes visual tokens through L_0 without driving a pathological signal along that route: the matched-compute MLP control of Section 4.2 passes A2 in isolation, a healthy backbone already routes early visual attention through L_0 , yet fails A1 and shows a null behavioral L_0 effect. Two natural negatives make the same point: UniTok and Emu3 both clear the A2 mass threshold yet fail A1, UniTok for lack of an early event and Emu3 for lack of propagation. Qwen-VQ makes the converse point, passing A1 and A3 and failing only the mass screen. Only the conjunction of a routing event (A2), a divergence signal that initiates early and propagates downstream (A1), and codebook off-manifold behavior (A3) recovers the correct verdict.

3.2 Constructive induction procedure with matched-compute control

The cross-architecture pattern across natural unified-VQ models admits a confounding explanation: those models differ from the continuous-projector models in many ways besides the tokenizer. To isolate vector quantization, we run a single-variable contrast. Starting from a healthy continuous-projector backbone (LLaVA-1.6 with Vicuna-7B and CLIP), we freeze the LM and the CLIP encoder and train *only* the projector under two configurations that differ in exactly one component. The first replaces the continuous CLIP+MLP projector with a VQ+Linear projector (a linear

map with codebook quantization at capacity K). The second replaces it with a 2-layer GeLU MLP, same as LLaVA’s original projector. Both configurations are trained for 2000 steps on the same 50k CC3M (Sharma et al., 2018) caption subset under identical hyperparameters (Appendix A.5); the only difference is whether the projector output is quantized through a codebook before reaching the LM.

If a brief projector-only training were itself sufficient to install the behavioral circuit, both configurations would do so. If the VQ bottleneck is the causally responsible component, only the VQ configuration should, and we treat this as the central causal test in Section 4.2. Two additional sweeps probe how the circuit depends on codebook structure. We vary capacity $K \in \{1024, 4096, 16384, 65536\}$ to test whether K alone predicts the circuit’s *polarity*, whether the model is biased toward YES or NO answers under the resulting circuit. We also train two finite-scalar-quantization FSQ (Mentzer et al., 2024) variants with different per-dimension level configurations (Appendix A.5). Because FSQ cannot exhibit codebook collapse by construction, the FSQ configuration tests whether the circuit forms in the absence of collapse. A second configuration was trained to distinguish a regime effect from a single-configuration artifact, but it fails A1 at initiation, so the comparison is between one circuit-installing configuration and one that is not.

3.3 Interventions and a sanity criterion

Interventions. We evaluate three mechanism-targeted interventions on VILA-U, each implemented as a forward-pass hook that modifies activations during inference without retraining: L_0 ablation, L_1 ablation, and VTI. L_ℓ ablation zeroes the self-attention output at layer- ℓ ; we report L_0 (the candidate circuit) and L_1 (the downstream writer in models such as Chameleon, Section 4.4). A separate window variant generalizes L_ℓ ablation to a contiguous range $[L_s, L_e]$ and is used only as a structural probe for Show-o and SEED-LLaMA, whose divergence has no single localized event to target, so no L_0 ablation is reported for them (Table 3). VTI (Liu et al., 2024c), described in Section 2, computes its steering direction from mean clean – corrupt residual differences over demonstration records; we add it at every decoder block with $\alpha = 0.005$. VTI is prior work rather than a contribution of ours; we include it as a mechanism-

adjacent comparison because its steering direction is built from the same clean – corrupt residual differences the diagnostic uses.

A sanity criterion for hallucination reduction.

An intervention that improves a binary accuracy number is not automatically a hallucination reduction. Liquid’s L_0 ablation, for instance, drives POPE yes-rate from 78 % to 0 % while producing all empty open-ended generations, the score improves because the model has stopped responding to the input, not because it has become better grounded. To rule out such cases, for each POPE record, we compute the logit gap $g = \text{logit}_{\text{YES}} - \text{logit}_{\text{NO}}$ and apply a *question-conditional* sanity criterion, computed separately on the yes-GT and no-GT record subsets, so a collapse affecting either alone is still detected. An intervention is *preserved* only if both (a) the ratio of baseline to intervention within-class standard deviation of g remains below 3 (the model still varies across inputs), and (b) the across-class mean difference $\bar{g}_{\text{yes-GT}} - \bar{g}_{\text{no-GT}}$ remains > 0.5 logits (the model still discriminates the two classes on average). An intervention failing either is a *degenerate emitter*: it demonstrates that the targeted circuit is present (Section 4.1) but not as a practical hallucination reduction.

3.4 Experimental setup

We summarize the setup here; full details are in Appendix A.5. We evaluate on POPE, AMBER, and CHAIR (Rohrbach et al., 2018) (500 images). For activation patching, we use 500 paired records (300 NaturalBench image_swap + 200 POPE gaussian_noise), each pair consisting of one YES and one NO record, which is what validates the paired-bootstrap CIs. A unified hook framework provides a consistent visual/question/answer/other token-category mapping across all models. All interval estimates reported in Appendix A.3 use 10,000 bootstrap resamples, resampling paired records on the binary benchmarks and images on CHAIR.

The studied models span natural VQ-VLMs (VILA-U (Wu et al., 2024), Chameleon-7B (Team, 2024), Liquid (Wu et al., 2025), Anole (Chern et al., 2024), Lumina-mGPT (Liu et al., 2024a), UniTok (Ma et al., 2025), Show-o (Xie et al., 2024), SEED-LLaMA (Ge et al., 2023), Emu3 (Wang et al., 2024), Janus-Pro (Chen et al., 2025)), continuous-projector controls (LLaVA-1.6 (Liu

Table 1: Five natural unified-VQ VLMs spanning three LLM families that pass the three-gate diagnostic. For $A1$, $init$ is the maximum relative divergence over layers $[0, 8)$ and $prop$ the maximum over $\ell \geq 12$; both must reach 0.4. $A2$ is scored on L_0 mass for projector-amplified models (P) and the $[0, 8)$ window total for unified-vocabulary models (U); both apply a ≥ 15 threshold (Section 3.1). Behavioral 95% CIs from 10,000 paired-record bootstrap resamples. [†]Lumina-mGPT is held out: baseline yes-rate at the behavioral floor.

Model	Backbone	Gate $A1$		$A2$ (tier)	$A3$	POPE yes-rate (%)	
		init $[0, 8)$	prop $\ell \geq 12$			base $\rightarrow$ L0	Δ [95% CI]
VILA-U-7B	Vicuna	0.932 (L_2)	0.724 (L_{16})	17.5 (P)	100	81.0 $\rightarrow$ 72.6	-8.4 [-9.7, -7.2]
Liquid-7B	Gemma	0.768 (L_1)	0.819 (L_{24})	19.2 (U)	100	78.2 $\rightarrow$ 0.0	-78.2 [-79.7, -76.7]
Chameleon-7B	Chameleon	0.635 (L_7)	7.267 (L_{31})	22.0 (U)	100	22.3 $\rightarrow$ 9.8	-12.5 [-14.3, -10.7]
Anole-7B	Chameleon	2.009 (L_1)	34.149 (L_{31})	111.2 (U)	100	75.3 $\rightarrow$ 2.0	-73.3 [-74.9, -71.7]
Lumina-mGPT-7B [†]	Chameleon	2.807 (L_6)	0.623 (L_{20})	175.0 (U)	100	0.07 $\rightarrow$ 0.0	-0.07 [-0.17, 0.00]

et al., 2024b), VILA (Lin et al., 2024), Qwen2.5-VL (Bai et al., 2025), HaploOmni (Xiao et al., 2025), LaViT (Jin et al., 2024), GILL (Koh et al., 2023)), and induced variants (LLaVA-VQ at $K \in \{1024, 4096, 16384, 65536\}$, two LLaVA-VQ-FSQ configurations, Qwen-VQ, and a matched-compute LLaVA-MLP variant). Detailed gate values for all models appear in Appendix A.1

4 Results

4.1 A shared early-layer routing circuit across VQ-tokenized VLMs

The diagnostic chain identifies a shared L_0 attention routing circuit across five natural unified-VQ VLMs spanning three LLM families (Table 1): Vicuna-7B in VILA-U, Gemma-7B (Team et al., 2024) in Liquid, and the Chameleon family in Chameleon-7B, Anole, and a held-out Lumina-mGPT. All five pass $A1$ (both halves), $A2$ under the two-tier criterion, and $A3$. Four exhibit large behavioral changes under L_0 ablation; Anole is particularly informative as an image-generation fine-tune of Chameleon-7B, with a gate- $A2$ window total five times larger than its base model, the circuit strengthens through generation fine-tuning, consistent with the broader claim that VQ-based training installs it. The layer-0 routing is a distributed pattern rather than a few-head circuit. Ranking the 32 heads at the routing layer by visual-to-prompt mass, thirteen heads are required to account for half of the total mass and twenty-two for eighty percent, with near-even per-head contributions (the strongest head carries 0.80 and the twenty-second still 0.63). No small subset dominates, which is why a full early-window knockout rather than a single-head ablation is required to disrupt the route (Section 4.4), and why UniTok’s diffuse layer-0

mass (Table 14) is disqualifying only in conjunction with its misaligned divergence event. “Circuit” here therefore denotes a coherent layer-level routing channel, not a sparse set of privileged heads.

Lumina-mGPT is held out: not in the threshold-calibration cohort. It passes all three gates with the strongest $A2$ signal observed in any model (Table 1), yet its behavioral change under L_0 ablation is null, its baseline yes-rate is 0.07%, leaving no room to drop. This structural-behavioral dissociation is itself informative: the diagnostic detects a structural property of the architecture independent of whether the language prior happens to manifest it behaviorally, and the held-out model confirms the thresholds were not over-fit to the calibration cohort. Under the conjunctive definition of $A1$, Lumina-mGPT’s binding value is its propagation maximum (0.623 at L_{20}) rather than its initiation peak, and that propagation value is the weakest among the natural models.

Polarity tracks codebook bias. The five models fall into three polarity regimes that align with their codebook’s training distribution. VILA-U and Liquid have universally YES-biased codebooks and YES-promoting circuits; Chameleon’s codebook is universally NO-biased and its baseline yes-rate is 22%. The induced FSQ model of Section 4.2, whose tokenizer cannot collapse by construction, produces a YES-suppressing circuit; a second FSQ configuration reproduces the polarity inversion behaviorally but does not install a gate-detectable early event. We frame this as associational rather than causal: Chameleon’s L_0 still mildly promotes YES despite the codebook’s NO-bias (its overall NO behavior comes from a downstream L_1 NO-writer, Section 4.4), and the LLaVA-VQ K -sweep is non-monotonic (Section 4.2).

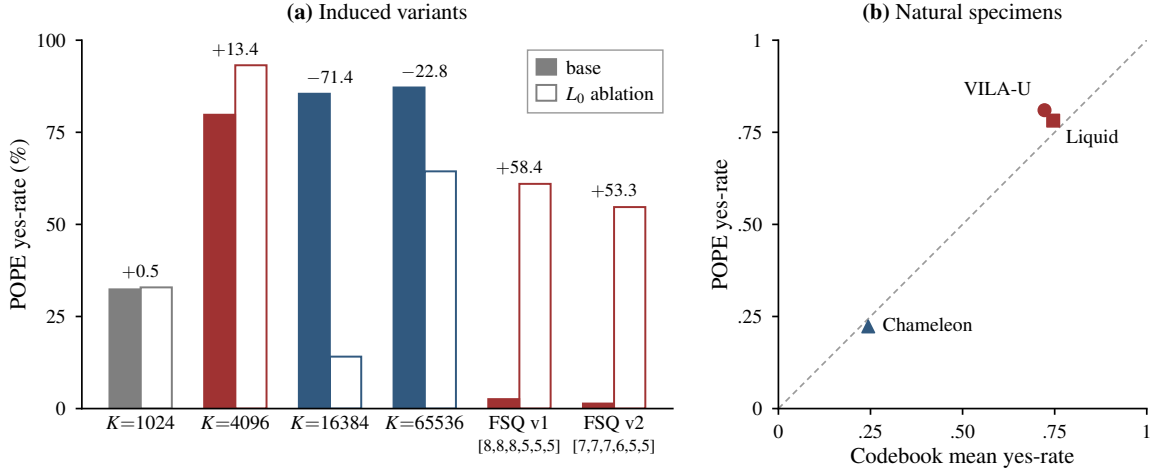


Figure 2: Codebook structure governs circuit polarity. (a) Induced LLaVA-VQ: capacity sweep and FSQ configurations, of which v1 installs the circuit and v2 does not. (b) Natural unified-VQ models: codebook mean yes-rate against behavioral baseline POPE yes-rate.

Table 2: The matched-compute contrast under identical training parameters on POPE. The VQ projector installs the L_0 routing circuit; the matched-compute MLP projector does not, failing both halves of A1, passing A2.

Metric	VQ+Linear ($K = 16384$)	MLP (matched-compute)
A1 init [0, 8)	1.276 ✓	0.206 ✗
A1 prop $\ell \geq 12$	1.117 ✓	0.106 ✗
A2 (L_0 mass)	26.02 ✓	25.52 ✓
yes-rate base $\rightarrow$ L0	85.5 $\rightarrow$ 14.1	13.7 $\rightarrow$ 8.7

4.2 Causal identification: induction and asymmetry

The cross-architecture pattern is consistent with VQ as the cause, but natural models differ in many ways besides their tokenizers. The single-variable contrast (Section 3.2) isolates the bottleneck: starting from LLaVA-1.6 with Vicuna-7B and CLIP frozen, we train one projector that quantizes through a codebook and one that does not, under identical hyperparameters.

The two configurations diverge cleanly (Figure 2). **LLaVA-VQ** installs the circuit: all three gates pass, and POPE yes-rate rises to 85.5 %, with L_0 ablation collapsing it to 14.1 %. The **matched-compute LLaVA-MLP control**, on identical data and budget, fails both halves of A1 (initiation 0.206; propagation 0.106) and shows only a negligible behavioral change under L_0 ablation (13.7% $\rightarrow$ 8.7%, a 5-point drop against the 71-point drop for LLaVA-VQ), yet passes A2 in isolation, with an L_0 mass of 25.52. Vicuna-7B’s pretrained backbone routes early visual attention through L_0 regardless of projector type; the chain-of-gates recovers the correct verdict where any single gate would

not.

Within the VQ configuration, sweeping $K \in \{1024, 4096, 16384, 65536\}$ shows that capacity alone does not determine polarity (Figure 2): all four collapse to a similar effective vocabulary, between roughly ~ 80 -160 active codes regardless of nominal capacity, yet their polarities differ: $K = 1024$ shows no measurable polarity shift, $K = 4096$ inverts to YES-suppressing, and the two larger capacities are YES-promoting. The $K = 4096$ inversion is not explained by the present account and is retained as an open anomaly (Section 6). Collapse is not the operative factor either. The FSQ configuration, which cannot collapse by construction, inverts polarity, yielding a YES-suppressing model. A second FSQ configuration reproduces that inversion behaviorally (POPE yes-rate +53.3 points, AMBER accuracy -19.8) but does not install a gate-detectable early divergence event: its initiation value of 0.379 falls below threshold, with divergence first clearing 0.4 only at L_{14} . The two lattices, therefore, agree on the behavioral consequence and differ on the mechanism, so the polarity inversion cannot be attributed to the

Table 3: Models the diagnostic rejects, each at a distinct gate. For A1 failures, the failing half is marked *init* or *prop*. L_0 Δ POPE is the accuracy change under L_0 ablation, where it was run; Show-o and SEED-LLaMA were probed with the window variant, so no L_0 value applies. Gates a model passes are discussed in Section 4.3.

Model	Class	Primary failure	L_0 Δ POPE	Note
<i>Continuous-projector controls (no routing concentration; A2 fails):</i>				
LLaVA-1.6-7B	continuous	A2	-0.7	Null ablation effect
Qwen2.5-VL-7B	continuous	A2	-34.3	Catastrophic collapse
HaploOmni-7B	continuous	A2	-16.1	Catastrophic collapse
LaVIT-7B	continuous	A2	-2.0	yes $\rightarrow$ no flip on AMBER
<i>VQ-tokenized models, distinct failure modes:</i>				
UniTok-7B	VQ + TokenEmb	A1 (init)	+1.9	Mass without early event
Show-o-1.3B	MAGVIT-v2 LFQ	A1 (init)	—	No divergence event
SEED-LLaMA-8B	VQ vocab-inj.	A1 (init)	—	Event too late for window
Emu3-Chat-8B	Emu3 VQ	A1 (prop)	-30.9	Early event, no propagation
Janus-Pro-7B	decoupled	A3	-0.4	No codebook to probe
<i>Induced negative controls:</i>				
Qwen-VQ	induced VQ	A2	-0.2	Collapse without circuit
LLaVA-MLP	matched MLP	A1 (init)	+0.3	Single-gate pass, chain rejects

early-layer circuit alone.

Induction is fast: 2000 steps install the circuit. Reversal is not. Training a continuous SigLIP MLP adapter on top of VILA-U’s VQ-pretrained Vicuna-7B for the same 2000 steps fails to remove the circuit; the adapted variant continues to pass A1 (0.672 initiation; 0.647 propagation) and A2, with L_0 mass rising from 17.50 to 20.53 and POPE yes-rate from 81 $\rightarrow$ 99.2%. It is nonetheless recorded as a chain FAIL, rejected at A3, because the continuous adapter leaves no codebook to probe; the structural gates it does pass are what carry the reversal argument. The asymmetry locates the circuit: in LLM backbone weights from VQ-based pretraining, not in the current projector. A healthy Vicuna-7B already routes L_0 attention with mass 25.52 (the MLP control above), so the routing pattern is in the backbone; what VQ pretraining installs is the pathological signal content flowing through that route, plus the residual divergence (A1) that propagates it forward.

4.3 The diagnostic generalizes across models

Beyond the models that pass, the diagnostic rejects fifteen models in the cohort, each for a mechanistically distinct reason; Table 3 details the eleven that most resemble the positive class. Continuous-projector VLMs (LLaVA-1.6, VILA, Qwen2.5-VL, HaploOmni, LaVIT) lack A2 concentration entirely; L_0 ablation is null on LLaVA-1.6 (-0.7 points on POPE) and catastrophic on Qwen2.5-VL (-34.3 points on POPE, -20.1 points on AMBER), an opposite-sign discrimination across architecture classes. VQ-tokenized models fail with distinct signatures. UniTok’s L_0 attention mass

(18.23) clears the A2 threshold, but it carries no early routing event: its early-window divergence reaches only 0.347 and first clears 0.4 at L_{13} , so it fails A1 at initiation, the opposite end from Emu3. Show-o’s divergence never approaches threshold, reaching 0.136 in the early window and a whole-stack maximum of 0.151, and so fails A1 at initiation. SEED-LLaMA fails A1 for the same reason but on a different curve: its divergence reaches only 0.278 within the early window and first clears 0.4 at L_{14} , outside the range A1-initiation measures. Emu3 passes A2 under its unified-vocabulary criterion (window mass $48.44 \geq 15$) and shows a genuine L_0 initiation event (early divergence 0.435), but it is the one model whose early spike fails to propagate: divergence decays to a downstream maximum of 0.215, so it fails A1 at propagation. Janus-Pro fails for a structural reason rooted in its decoupled design: its understanding pathway uses continuous SigLIP features with no VQ codebook to probe, so it fails A3 outright; its L_0 mass of 13.57 is incidentally also below the projector-amplified threshold. The Qwen-VQ induced control is especially clean: its codebook collapses severely yet $A2 = 6.07$, and POPE collapses to all-NO, establishing that codebook collapse alone is not sufficient to install the circuit. Conversely, the FSQ configuration of Section 4.2 installs the circuit without any collapse, establishing that collapse is not necessary either, though on a single model, since the second FSQ configuration fails A1. Both observations together point to the VQ bottleneck, discretization through a codebook, as the operative factor, with the surviving codes’ training distribution governing polarity.

Table 4: Mechanism-targeted interventions. **(a)** Three interventions on VILA-U, with paired-bootstrap 95% CIs reported in Table 9. **(b)** Architecture-specificity of L_0 ablation: the same intervention applied across the cohort. Each cell is tagged by outcome, where null denotes a 95% paired-bootstrap CI spanning zero and *catas.* denotes a catastrophic collapse of the question-conditional logit gap (Section 3.3).

Method / Model	POPE acc Δ	AMBER acc Δ
<i>(a) Mechanism-targeted interventions on VILA-U:</i>		
L_0 ablation	+4.23	+16.9
L_1 ablation	+11.60	+23.37
VTI ($\alpha=0.005$)	+8.3	+30.23
<i>(b) Architecture-specificity of L_0 ablation (same intervention, different models):</i>		
VILA-U	+4.23 (restored)	+16.9 (restored)
Chameleon-7B	−1.4 (null)	−4.4 (degraded)
LLaVA-1.6	−0.7 (null)	−1.0 (degraded)
Qwen2.5-VL	−34.3 (catas.)	−20.1 (catas.)
Emu3-Chat	−30.9 (catas.)	−30.2 (catas.)

Three single-measure A_2 alternatives each produce 2-3 misclassifications on our 10-model cohort (Appendix A.6); the two-tier rule instead derives from the σ -calibration tier structure (Section 3.1), an architectural property documented before any behavioral outcome was measured.

4.4 Mechanism-targeted intervention reduces hallucination in generation

The mechanism provides a targeted intervention: ablate L_0 (or L_1 when the writer is downstream, as in Chameleon). On VILA-U, the model that passes the sanity criterion cleanly, all three mechanism-targeted interventions yield substantial gains with tight paired-bootstrap CIs (Table 4, top): L_0 ablation reaches +4.23 POPE; L_1 ablation, targeting the downstream amplifier in the writer/reader pattern, +11.60 POPE and +23.37 AMBER; VTI ($\alpha = 0.005$) gives the largest AMBER effect at +30.23. The writer/reader pattern generalizes across architectures with codebook-bias-tracking signs: VILA-U has YES-promoting L_0 and L_1 , Chameleon a mildly YES-promoting L_0 alongside a dominant L_1 NO-writer (L_1 ablation flips POPE yes-rate 22.3 $\rightarrow$ 66.2 %), and Liquid a dominant L_0 with a partial L_1 . Architecture-specificity is sharp (Table 4, bottom): the same L_0 ablation is null on LLaVA-1.6 and catastrophic on Qwen2.5-VL, opposite-sign discrimination hard to explain on any non-mechanistic account.

The benefit is also specific to the early routing layers rather than a generic consequence of perturbing the network. Applying the same ablation layer by layer, it helps at layers 0 and 1 (+4.23 and

+11.60 on POPE) but worsens accuracy at layers 2 and 4 (−10.67 and −4.47) and has no effect at layers 8 and 16; a generic-degradation account predicts no such sign structure. Scaling the routing layer’s attention output rather than zeroing it reveals a graded dose-response, with attenuation factors of 0.75, 0.50, and 0.25 yielding +8.67, +8.17, and +12.30 points, so a soft intervention outperforms full ablation on binary accuracy. Because the remainder of this section shows that binary gains need not transfer to open-ended generation, the attenuation sweep is reported here as a specificity control; the generative comparison under graded attenuation is left to future work.

Would a mechanism-agnostic method, given the same tuning budget, do equally well? We swept VCD over 16 configurations and DoLA over 20 on POPE and AMBER (Appendix A.4). On binary benchmarks **tuned DoLA wins**: best DoLA ($\alpha = 0.10$, $\ell = 16$) reaches POPE 0.794 and AMBER 0.795, beating our best mechanism-targeted intervention by +5.9 and +9.6 points respectively. On a single-token forced choice, a tuned decoding-time logit fix can match or exceed mechanism-targeted ablation.

The binary picture is not the whole picture. On open-ended COCO captioning via CHAIR, with the winning sweep configurations (Table 5), the ranking reverses: **only L_0 ablation reduces CHAIR_i** (10.3 $\rightarrow$ 7.1 %, −31 % relative), while tuned DoLA barely moves it (−0.4 %) and tuned VCD *worsens* it, collapsing recall to 43.9 %. This reduction is not an artifact of shorter captions: truncating each

Table 5: CHAIR open-ended captioning on 500 COCO (Lin et al., 2014) images. AvgLen uses the CHAIR-scorer’s tokenization.

Condition	CHAIR _s	CHAIR _i	Recall	AvgLen
<i>VILA-U:</i>				
Baseline	35.8	10.3	58.5	158
L_0 ablation (ours)	17.6	7.1	51.6	78
VCD ($\alpha=0.50, \sigma=80$)	20.6	10.7	43.9	84
DoLA ($\alpha=0.10, \ell=16$)	31.8	9.9	55.4	156
<i>LLaVA-VQ $K=65,536$ (induced, same backbone as VILA-U):</i>				
Baseline	31.2	52.5	11.1	158
L_0 ablation (ours)	19.6	42.7	9.5	145

baseline caption to the length of its paired ablated caption lowers baseline CHAIR_i only from 10.3 to 9.53, so brevity accounts for less than a third of the observed drop. The length-controlled reduction attributable to the intervention is 2.40 points, holding consistently across captions with different object counts. Image-level intervals for all three metrics are in Appendix A.3.

The dissociation is mechanistically interpretable: DoLA’s early-late layer contrast calibrates a forced choice but does not disrupt the L_0 circuit that injects hallucinated objects during generation; VCD’s noise-subtraction penalizes object tokens broadly, cutting recall without reducing the hallucination rate among emitted objects. The reduction also preserves substantial informativeness (Table 15): object mentions per caption fall modestly ($6.02 \rightarrow 5.38$) and object recall from 58.5 to 51.6, while reference alignment measured by METEOR (?) improves ($0.127 \rightarrow 0.177$). Hallucination is reduced while captions still name about five objects and recall roughly half of the ground-truth objects.

A second CHAIR run on the induced **LLaVA-VQ** $K=65,536$ variant strengthens the end-to-end story: baseline CHAIR_i is 52.5 %, five times VILA-U’s, so the controlled architectural change inflates open-ended hallucination fivefold on the same backbone, and L_0 ablation cuts it to 42.7 % (-19% relative) with only 8 % length reduction. At the image level, however, the induced model’s CHAIR_i reduction is not significant ($-9.71 [-30.40, +11.01]$); its CHAIR_s reduction is, and its length-matched CHAIR_i difference is -5.82 points. The induced variant is therefore supporting but underpowered evidence, with the VILA-U result as the clean, significant one. The

intervention’s effect is concentrated on exactly the images that hallucinate most: across the 500 images, the correlation between baseline hallucination count and the change under ablation is -0.383 , and images with at least two baseline hallucinations (76 images) lose on average 1.59 hallucinated objects while images with at most one (424 images) are essentially unchanged ($+0.01$). The circuit thus acts where hallucination actually occurs, leaving grounded captions intact. This induction-to-intervention sequence remains the most direct end-to-end link between the mechanism and the behavior.

5 Discussion

Implications for VQ tokenizer design. The polarity association has practical implications. In practice, unified VQ tokenizers tend to develop codebook collapse, and the natural models we examined that do collapse (VILA-U, Liquid, Chameleon, Anole) all carry the L_0 routing circuit; whether the model hallucinates by over-claiming or by denying then depends on what the training distribution encoded into surviving codes. The induced Qwen-VQ counterexample shows that collapse alone is insufficient; the circuit requires both the VQ bottleneck and a backbone where it can form, while the non-collapsing FSQ configuration of Section 4.2 installs the same circuit with inverted polarity, so collapse is neither necessary nor sufficient. The L_0 routing structure itself is the problem, regardless of the polarity sign.

Where the circuit lives. The reverse-ablation failure indicates that the L_0 circuit is in the LLM backbone weights, not the projector. The matched-compute LLaVA-MLP control reinforces this: even a healthy Vicuna-7B backbone with an MLP pro-

jector trained for 2000 steps shows L_0 attention mass 25.52, the *routing pattern* is in the backbone already. What VQ pretraining installs is the pathological signal content that flows through that routing pattern, plus the residual divergence that propagates it forward.

Binary vs. generative dissociation. The cleanest scientific finding is the dissociation between binary and generative metrics. Tuned DoLA wins POPE/AMBER; only L_0 ablation wins CHAIR. This is exactly what a mechanistic account predicts, and a decoding-time fix does not: routing circuits operate during open-ended generation, not just at single-token forced-choice. This link is supported directly by the per-image finding that the intervention’s benefit scales with an image’s baseline hallucination (Section 4.4). The dissociation makes the mechanism-targeted intervention non-redundant with the mechanism-agnostic baselines, even when those baselines win on a single benchmark.

6 Conclusion

We identified an early-layer attention routing circuit shared across VQ-tokenized vision-language models, validated it on a held-out model, isolated its cause via a single-variable construction with a matched-compute control, and showed that ablating it is the only method (in a 36-config sweep of tuned baselines) that reduces object hallucination in open-ended generation. The circuit is architecturally caused, lives in LLM backbone weights from VQ-pretraining, and is discriminable by a three-gate diagnostic that requires no single-measure fit. The findings reframe object hallucination in unified VLMs as a property of *architecture and pretraining*, not of decoding-time calibration, and point to VQ tokenizer design as the leverage point for the next generation of unified VLMs.

Limitations

Several limitations temper our results. First, our reverse-ablation result is reported as an asymmetry at matched budget, not as impossibility: a longer fine-tuning run with the LM unfrozen might, in principle, reverse the circuit, and we make only the weaker claim. Second, the polarity association in Section 4.1 is associational, not causal; two anomalies (the $K = 4096$ inversion and Chameleon’s mildly YES-promoting L_0) lack a mechanistic explanation, and we do not claim that intervening on

codebook bias would flip polarity. Third, while the circuit *topology* is shared across architectures, the intervention direction is model-specific: per-model VTI directions have pairwise cosine similarity ≤ 0.054 and must be re-derived per architecture. Fourth, the sanity criterion of Section 3.3 excludes Liquid and Anole as practical intervention models (their L_0 ablation collapses to a degenerate emitter); we use them as gate-level evidence only. Relatedly, the distributed-head account of the L_0 routing rests on the per-head mass distribution, so the head-level claim is structural rather than causal. Further, the claim that codebook collapse is not necessary for the circuit rests on a single FSQ model: a second FSQ configuration reproduces the polarity-inversion behavior but fails A1 at initiation (0.379), so the two lattices agree on the behavioral consequence and differ in the mechanism. Finally, the open-ended (CHAIR) result rests on VILA-U together with a single induced variant, so the generative intervention is evidenced more narrowly than the diagnostic itself; the induced variant’s CHAIR_i reduction is not significant at the image level, so the significant open-ended evidence rests on VILA-U alone.

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A Appendix

A.1 Cohort taxonomy

Tables 6 and 7 give the complete model cohort. Table 6 reports architecture, σ -calibration, and the three structural gate values; Table 7 reports behavioral results under L_0 ablation. A_2 is two-tier: projector-amplified models require L_0 mass ≥ 15 while unified-vocabulary models require $[0, 8)$ window total ≥ 15 .

A.2 σ calibration table

Table 8 reports per-model σ calibration values and achieved cosine similarities for the causal sweep.

A.3 Paired-bootstrap CIs

Table 9 reports all 24 record-level paired-bootstrap 95% confidence intervals for the binary benchmarks (10,000 resamples per cell). Every interval in Table 9 is at most 3.8 pp wide; AMBER intervals are at most 1.7 pp wide, owing to the larger record count (14,216 for AMBER vs. 3,000 for POPE). Table 10 reports the corresponding intervals for open-ended captioning. These resample the 500 COCO images rather than paired records, since CHAIR is scored per caption, and are correspondingly wider: CHAIR_i falls from 10.30 [8.80, 11.82] to 7.13 [5.04, 9.57], a difference of -3.17 [-5.37, -0.76] whose interval excludes zero, while CHAIR_s falls by 18.20 [13.80, 22.80] and recall by 6.89 [4.86, 8.98].

A.4 Sweep grid

Tables 11 and 12 give accuracy on POPE-adversarial and AMBER for all 36 hyperparameter configurations swept. VILA-U baseline: POPE acc = 0.619; AMBER acc = 0.397. Winning configurations are **bold**.

A.5 Training hyperparameters

Table 13 lists the projector configuration and final codebook utilization for all induced models. All were trained for 2,000 steps on the same 50k-caption CC3M subset with batch size 8, gradient accumulation 4, and learning rate $2e-3$, with the CLIP ViT-L/14 encoder and the Vicuna-7B language model frozen throughout. “Codes” is the number of distinct codebook entries with positive EMA usage at the final checkpoint.

A.6 Three-gate diagnostic: alternative single-measure A_2 criteria

Table 14 reports the four A_2 mass metrics for the ten-model subset used to motivate the two-tier rule. Three single-measure alternatives are evaluated against the correct two-tier A_2 mass verdict: (i) a uniform L_0 mass threshold of 15 produces three errors: Liquid, Chameleon and Emu3, which carry sufficient mass at their architecture-class window but fall below a uniform L_0 threshold; (ii) a uniform $[0, 8)$ window-mass threshold produces two false positives, Qwen-VQ and Janus-Pro, whose window totals clear 15 despite negligible early-layer mass; (iii) a normalized window measure ($[0, 8)/\text{total mass}$) produces at best two errors, both false positives, at any threshold. No single mass threshold reproduces the architecture-class verdicts. The two-tier rule is grounded in the architectural σ -tier distinction: projector-amplified vs. unified-vocab is a property of the model’s embedding pathway, not a post-hoc fit to gate values.

A direct sweep of the thresholds themselves confirms the same conclusion. Over the nineteen cohort models carrying instrumented gate values (ten positive, nine negative), holding the diagnostic definition fixed and sweeping one gate at a time, the partition is invariant to A_3 across 60-95% with zero verdict changes. The A_2 partition is invariant from 8 – 15 and first changes only at 18, where the lowest-mass positive model (mass 17.50) drops out, so the binding constraint is the weakest true positive rather than a tuned cutoff. A_1 is tightly bound on both sides: lowering the threshold to 0.35 admits FSQ v2 (initiation 0.379) and 0.30 admits UniTok (0.347), while raising it to 0.45 drops the FSQ v1 positive. The default is the only value in the swept range that reproduces the partition, constrained from below by two negatives and from above by a positive. The remaining six cohort models carry no measurable gate values and are trivial rejects.

A.7 Open-ended generation: supporting analyses

By category, total hallucinated mentions fall from 310 to 192, a 38% reduction; 36 categories improve, 11 worsen, and 31 are unchanged. The most-hallucinated categories are strongly suppressed (orange 41 $\rightarrow$ 11, chair 34 $\rightarrow$ 8, bird 12 $\rightarrow$ 1, bear 10 $\rightarrow$ 0), while a few scene-typical categories rise (dining table 20 $\rightarrow$ 43, sports ball 8 $\rightarrow$ 25), indi-

Table 6: Full cohort: architecture, σ -calibration, and the three structural gate values. σ_{cal} is calibrated to clean-noisy cosine similarity ≈ 0.5 . $A1$ is conjunctive (init over $[0, 8)$, prop over $\ell \geq 12$; both must reach 0.4). $A2$ is the visual $\rightarrow$ prompt-last attention mass at each model’s architecture-class window (L_0 sink for projector-amplified and decoupled models, $[0, 8)$ total for unified-vocab), and is not directly comparable across windows; $A3$ is the VQ off-manifold rate. Em-dash entries denote not applicable or not measured. Verdict combines the structural gate chain with behavioral polarity under L_0 ablation. Backbone families are counted by base pretrained LLM.

Model	Class	Backbone	σ_{cal}	Gate $A1$		$A2$	$A3$	Verdict
				init $[0, 8)$	prop $\ell \geq 12$			
<i>Natural PASS models</i>								
VILA-U-7B	Nat. VQ collapsed	Vicuna-7B	1.0	0.932 (L_2)	0.724 (L_{16})	17.50	100%	PASS
Liquid-7B	Nat. VQ collapsed	Gemma-7B	0.01	0.768 (L_1)	0.819 (L_{24})	19.19	100%	PASS — degenerate L_0
Chameleon-7B	Nat. VQ collapsed	Chameleon-7B	0.0063	0.635 (L_7)	7.267 (L_{31})	22.04	100%	PASS
Anole-7B	Nat. VQ + FT	Chameleon-7B	0.0063	2.009 (L_1)	34.149 (L_{31})	111.20	100%	PASS — degenerate L_0
Lumina-mGPT-7B	Nat. VQ	Chameleon-7B	0.0075	2.807 (L_6)	0.623 (L_{20})	175.02	100%	PASS — structural; behavioral floor
<i>Induced PASS models</i>								
LLaVA-VQ-K1024	Induced K-sweep	Vicuna-7B	2.0	1.187 (L_6)	1.125 (L_{12})	25.81	100%	PASS — induced, no behav. effect
LLaVA-VQ-K4096	Induced K-sweep	Vicuna-7B	2.0	1.140 (L_6)	1.029 (L_{12})	26.10	100%	PASS — induced, inverted (anomaly)
LLaVA-VQ-K16384	Induced VQ coll.	Vicuna-7B	2.0	1.276 (L_5)	1.117 (L_{12})	26.02	100%	PASS — induced
LLaVA-VQ-K65536	Induced K-sweep	Vicuna-7B	2.0	1.228 (L_5)	1.102 (L_{12})	25.89	100%	PASS — induced, YES-promoting (weak)
LLaVA-VQ-FSQ v1	Induced no-coll.	Vicuna-7B	0.5	0.450 (L_7)	0.466 (L_{12})	27.00	—	PASS — induced, inverted polarity
<i>FAIL models</i>								
LLaVA-VQ-FSQ v2	Induced no-coll.	Vicuna-7B	2.0	0.379 (L_7)	0.510 (L_{14})	26.91	—	FAIL — $A1$ initiation
LLaVA-MLP (matched)	Induced no-VQ	Vicuna-7B	0.5	0.206 (L_2)	0.106 (L_{25})	25.52	—	FAIL — $A1$ initiation + null behav.
UniTok	Nat. VQ + TokenEmb	Vicuna-7B	0.2	0.347 (L_2)	0.493 (L_{13})	18.23	100%	FAIL — $A1$ initiation
Show-o-1.3B	Nat. VQ vocab-inj	Phi-1.5	0.1	0.136 (L_6)	0.147 (L_{15})	5.838	100%	FAIL — $A1$ initiation
SEED-LLaMA-8B	Nat. VQ vocab-inj	LLaMA-2-7B	0.033	0.278 (L_7)	0.409 (L_{14})	9.455	86.3%	FAIL — $A1$ initiation
Emu3-Chat	Nat. VQ no proj	Emu3	0.01	0.435 (L_0)	0.215 (L_{28})	48.44	100%	FAIL — $A1$ propagation
Qwen-VQ	Induced VQ coll.	Qwen2.5-7B	1.0	0.468 (L_7)	0.687 (L_{17})	6.07	100%	FAIL — $A2$, collapse w/o circuit
Janus-Pro-7B	Decoupled cts. und.	DeepSeek LLM	7.0	0.786 (L_5)	0.876 (L_{12})	13.57	—	FAIL — $A3$ (no codebook); $A2$ sub-thres.
VILA-U-MLP (reverse)	Adapted cts.	Vicuna-7B	10.0	0.672 (L_7)	0.647 (L_{13})	20.53	—	FAIL — $A3$ (no codebook); reversal control
LLaVA-1.6	Continuous (ctrl)	Vicuna-7B	0.5	—	—	—	—	FAIL — continuous, no circuit
Qwen2.5-VL-7B-Instr	Continuous (ctrl)	Qwen2.5-7B	0.5	—	—	—	—	FAIL — continuous, no circuit
HaploOmni	Continuous (ctrl)	Qwen2.5-7B	0.5	—	—	—	—	FAIL — continuous, no circuit
LaVIT-7B	Continuous (ctrl)	LLaMA-2-7B	0.5	—	—	—	—	FAIL — continuous, no circuit
VILA-7B	Continuous (ctrl)	LLaMA-2-7B	0.5	—	—	—	—	FAIL — continuous, no circuit
GILL	Continuous CLIP+OPT	OPT-6.7B	10.0	—	—	—	—	FAIL — continuous, OPT-norm artifact

cating that suppression is broad and circuit-level rather than a per-code edit, with some residual shift toward scene-generic objects.

CIDeR (Vedantam et al., 2015) is not reported, as it sits near zero for both conditions when detailed captions of roughly 158 words are scored against short COCO references.

A.8 Robustness to corruption type

Section 3.1 reports that the $A1$ divergence is not specific to Gaussian embedding noise. This appendix gives the per-corruption detail behind that claim, measured on one model from each VQ tier. Four corruptions were compared: Gaussian embedding noise (the default), token dropout, visual-token permutation, and image blur applied before encoding. The first two reproduce the early peak; the latter two qualify rather than confirm the result. On VILA-U (projector-amplified) and Chameleon (unified-vocabulary), token dropout, zeroing half the visual tokens, strengthens the early peak (VILA-U $0.93 \rightarrow 1.08$ at layer 2; Chameleon $0.64 \rightarrow 2.25$ at layer 1), in each case within the initiation window. VILA-U’s dropout peak coincides with its Gaussian initiation maximum; Chameleon’s shifts earlier within the window. Image blur surfaces the early peak only in the unified-vocabulary model, whose full-image tokenizer is blur-sensitive, and

not in the projector-amplified model, whose tokenizer is blur-robust. The precise claim is therefore that the circuit responds to corruption of visual content and is order-invariant, rather than that any input perturbation triggers it.

Table 7: Full cohort: behavioral results under L_0 ablation, for every model on which an L_0 ablation was run. Δacc and Δyr (yes-rate) are percentage-point changes. Models probed only with the window variant, and continuous controls not ablated, are omitted. ^d generation-confirmed degenerate emitter (whitespace output): Δyr is real, but question-conditionality is lost. ^c catastrophic collapse. For the collapsed-baseline induced models, POPE Δacc is insensitive (base acc $\approx$ chance); the circuit shows in Δyr and AMBER Δacc .

Model	POPE			AMBER		
	Base Acc	Δacc	Δyr	Base Acc	Δacc	Δyr
<i>PASS models</i>						
VILA-U	0.619	+4.2	-8.4	0.397	+16.9	-19.3
Liquid ^d	0.650	-15.0	-78.2	0.646	+1.7	-51.7
Chameleon	0.502	-1.4	-12.5	0.639	-4.4	+12.6
Anole ^d	0.659	-16.0	-73.3	0.502	+15.6	-58.8
Lumina-mGPT	0.499	+0.1	-0.1	0.663	+0.1	-0.1
LLaVA-VQ-FSQ v1	0.498	+0.8	+58.4	0.647	-20.5	+40.8
LLaVA-VQ-K1024	0.486	+1.1	+0.5	0.529	-7.6	+5.2
LLaVA-VQ-K4096	0.503	-1.3	+13.4	0.392	-3.4	-5.0
LLaVA-VQ-K16384	0.530	+0.1	-71.4	0.402	+16.7	-66.9
LLaVA-VQ-K65536	0.494	+3.7	-22.8	0.421	-1.2	-16.1
<i>Non-PASS and control models</i>						
LLaVA-VQ-FSQ v2	0.503	+7.6	+53.3	0.663	-19.8	+41.2
LLaVA-MLP ctrl	0.501	+0.3	-5.0	0.635	+1.4	-3.5
LLaVA-1.6	0.860	-0.7	-1.0	0.814	-1.0	+0.4
Qwen2.5-VL ^c	0.847	-34.3	-35.9	0.841	-20.1	-16.8
Emu3-Chat ^c	0.791	-30.9	-10.6	0.831	-30.2	+2.5
HaploOmni ^c	0.830	-16.1	-4.5	0.761	-12.9	-8.6

Table 8: Per-model noise standard deviation σ_{cal} for the causal sweep. Calibrated so the cosine similarity between clean and Gaussian-corrupted post-projector embeddings is approximately 0.5.

Model	σ_{cal}	Achieved cos-sim	Tier
VILA-U	1.0	0.39	Projector-amplified VQ
LLaVA-VQ-K16384	2.0	0.40	Projector-amplified VQ
LLaVA-VQ-FSQ v1	0.5	0.57	Projector-amplified VQ (FSQ)
LLaVA-VQ-FSQ v2	2.0	0.59	Projector-amplified VQ (FSQ)
LLaVA-1.6 / VILA / Qwen2.5-VL / HaploOmni / LaVIT	0.5	≈ 0.50	Continuous projector
Janus-Pro	7.0	0.50	Continuous (SigLIP, large norms)
GILL	10.0	0.49	Continuous (OPT norm artifact)
UniTok	0.2	0.66	VQ + TokenEmbedder
Show-o	0.1	0.41	Unified-vocab VQ (vocab-injection)
SEED-LLaMA	0.033	0.49	Unified-vocab VQ (vocab-injection)
Emu3-Chat	0.01	0.51	Unified-vocab VQ (no projector)
Chameleon	0.0063	0.48	Unified-vocab VQ
Liquid	0.01	0.50	Unified-vocab VQ
Lumina-mGPT	0.0075	0.50	Unified-vocab VQ
Anole	0.0063	0.48	Unified-vocab VQ
LLaVA-MLP (matched ctrl)	0.5	0.49	Continuous induced (control; no noise)
VILA-U-MLP (reverse)	10.0	0.51	Continuous adapter on VQ backbone
Qwen-VQ	1.0	0.55	Projector-amplified VQ (induced)

Table 9: Full 10 000-resample paired-bootstrap 95% CIs. Δ and CI bounds are in percentage-point units of the reported metric.

Condition	Bench	Metric	Δ (pp)	CI lo	CI hi	Width
VILA-U L_0 ablation	POPE	acc	+4.23	+2.90	+5.53	2.63
VILA-U L_0 ablation	POPE	yes_rate	−8.43	−9.73	−7.17	2.57
VILA-U L_0 ablation	AMBER	acc	+16.92	+16.26	+17.62	1.36
VILA-U L_0 ablation	AMBER	yes_rate	−19.26	−19.93	−18.61	1.32
VILA-U L_1 ablation	POPE	acc	+11.60	+9.70	+13.47	3.77
VILA-U L_1 ablation	POPE	yes_rate	−28.73	−30.40	−27.10	3.30
VILA-U L_1 ablation	AMBER	acc	+23.37	+22.59	+24.16	1.56
VILA-U L_1 ablation	AMBER	yes_rate	−26.73	−27.48	−26.00	1.48
VILA-U VTI ($\alpha=0.005$)	AMBER	acc	+30.23	+29.42	+31.05	1.63
VILA-U VTI ($\alpha=0.005$)	AMBER	yes_rate	−31.77	−32.58	−30.97	1.61
Liquid L_0 ablation	POPE	acc	−15.03	−18.13	−11.97	6.17
Liquid L_0 ablation	POPE	yes_rate	−78.17	−79.67	−76.70	2.97
Liquid L_0 ablation	AMBER	acc	+1.72	+0.55	+2.91	2.36
Liquid L_0 ablation	AMBER	yes_rate	−51.74	−52.55	−50.89	1.66
Liquid L_1 ablation	POPE	acc	+1.53	−0.13	+3.13	3.27
Liquid L_1 ablation	POPE	yes_rate	−14.80	−16.37	−13.23	3.13
Liquid L_1 ablation	AMBER	acc	+0.35	−0.37	+1.09	1.46
Liquid L_1 ablation	AMBER	yes_rate	−9.85	−10.54	−9.14	1.39
Chameleon L_0 ablation	POPE	acc	−1.37	−3.30	+0.53	3.83
Chameleon L_0 ablation	POPE	yes_rate	−12.50	−14.33	−10.67	3.67
Chameleon L_0 ablation	AMBER	acc	−4.39	−5.19	−3.59	1.60
Chameleon L_0 ablation	AMBER	yes_rate	+12.63	+11.87	+13.41	1.54
Chameleon L_1 ablation	POPE	acc	+1.33	−1.47	+4.17	5.63
Chameleon L_1 ablation	POPE	yes_rate	+43.87	+41.43	+46.27	4.83
Chameleon L_1 ablation	AMBER	acc	−16.74	−17.88	−15.55	2.33
Chameleon L_1 ablation	AMBER	yes_rate	+47.17	+46.26	+48.09	1.82

Table 10: Image-level paired-bootstrap 95% confidence intervals for open-ended captioning (500 COCO images, 10,000 resamples). For the primary model (VILA-U), both hallucination reductions are significant, as is the accompanying recall cost. For the induced LLaVA-VQ ($K = 65,536$) model, the CHAIR_s and Recall differences are significant, but the CHAIR_i difference is not; the severely collapsed recall ($\sim 11\%$) makes CHAIR_i suggestive only.

Metric	Baseline	L_0 ablation	Δ [95% CI]
<i>VILA-U:</i>			
CHAIR _i	10.30 [8.80, 11.82]	7.13 [5.04, 9.57]	−3.17 [−5.37, −0.76]
CHAIR _s	35.8 [31.6, 40.0]	17.6 [14.2, 21.0]	−18.20 [−22.80, −13.80]
Recall	58.5 [55.69, 61.35]	51.6 [48.75, 54.53]	−6.89 [−8.98, −4.86]
<i>Induced LLaVA-VQ ($K = 65,536$):</i>			
CHAIR _i	52.46 [41.02, 63.57]	42.74 [24.73, 59.76]	−9.71 [−30.40, +11.01]
CHAIR _s	31.2 [27.4, 35.4]	19.6 [16.2, 23.2]	−11.60 [−16.20, −7.00]
Recall	11.14 [9.80, 12.56]	9.54 [8.33, 10.77]	−1.60 [−3.00, −0.24]

Table 11: VCD sweep: POPE acc / AMBER acc at each (α, σ) combination.

α	$\sigma = 10$		$\sigma = 20$		$\sigma = 40$		$\sigma = 80$	
	POPE	AMBER	POPE	AMBER	POPE	AMBER	POPE	AMBER
0.50	0.721	0.698	0.724	0.698	0.723	0.703	0.729	0.707
1.00	0.504	0.528	0.543	0.545	0.604	0.592	0.656	0.646
1.50	0.312	0.320	0.324	0.334	0.363	0.628	0.456	0.640
2.00	0.305	0.311	0.311	0.518	0.345	0.630	0.396	0.640

Table 12: DoLA sweep: POPE acc / AMBER acc at each (α, ℓ) combination.

α	$\ell = 0$		$\ell = 4$		$\ell = 8$		$\ell = 16$		$\ell = 24$	
	POPE	AMBER	POPE	AMBER	POPE	AMBER	POPE	AMBER	POPE	AMBER
0.10	0.714	0.694	0.714	0.694	0.716	0.697	0.794	0.795	0.481	0.336
0.30	0.714	0.694	0.709	0.686	0.716	0.695	0.648	0.718	0.196	0.219
0.50	0.714	0.694	0.704	0.675	0.715	0.693	0.500	0.664	0.202	0.223
1.00	0.709	0.686	0.684	0.651	0.714	0.687	0.497	0.650	0.205	0.225

Table 13: Projector configuration and final codebook utilization for all induced models.

Model	Projector	K	Codes
LLaVA-VQ-K1024	VQ+Linear	1,024	158 (15.4%)
LLaVA-VQ-K4096	VQ+Linear	4,096	95 (2.3%)
LLaVA-VQ-K16384	VQ+Linear	16,384	82 (0.5%)
LLaVA-VQ-K65536	VQ+Linear	65,536	85 (0.13%)
LLaVA-VQ-FSQ v1	FSQ+Linear	64,000	full
LLaVA-VQ-FSQ v2	FSQ+Linear	51,450	full
Qwen-VQ	VQ+Linear	16,384	33 (0.2%)
LLaVA-MLP (ctrl)	2-layer GeLU MLP	n/a	n/a

Table 14: $A2$ mass metrics for the ten-model subset. Tier is the architectural class from σ calibration; $A2$ (mass) is the two-tier structural screen (Section 3.1); Chain is the overall verdict with its rejecting gate. The mass screen tests only whether early-layer attention mass is present, not whether it forms an aligned routing circuit, that is, $A1$. [†]VILA-U-MLP is the reverse-ablation variant, not a fresh model. [‡]UniTok clears the mass screen, but its mass sits a dozen layers from the divergence event at L_{13} , so it passes without an aligned circuit (rejected at $A1$ -initiation).

Model	Tier	$A2$ (mass)	Chain	L_0 mass	$[0, 8)$ mass	Total mass	Norm. $[0, 8)$	Note
<i>$A2$ (mass) PASS</i>								
VILA-U	proj-amp	PASS	PASS	17.50	24.17	121.41	0.199	Natural, Vicuna-7B
LLaVA-VQ-K16384	proj-amp	PASS	PASS	26.02	62.54	199.91	0.313	Induced $K=16384$
Liquid	uni-vocab	PASS	PASS	2.47	19.19	75.31	0.255	Gemma-7B; L_0 low by design
Chameleon	uni-vocab	PASS	PASS	7.55	22.04	187.71	0.117	Unified-vocab tier
Anole	uni-vocab	PASS	PASS	27.08	111.20	504.12	0.221	FT Chameleon; $5 \times$ base
UniTok [‡]	proj-amp	PASS	FAIL ($A1$)	18.23	31.62	59.22	0.534	L_0 mass; event at L_{13}
VILA-U-MLP [†]	proj-amp	PASS	FAIL ($A3$)	20.53	134.73	543.00	0.248	Reverse ablation
Emu3-Chat	uni-vocab	PASS	FAIL ($A1$)	12.50	48.44	115.41	0.420	L_0 init.; no propagation
<i>$A2$ (mass) FAIL</i>								
Qwen-VQ	proj-amp	FAIL	FAIL ($A2$)	6.07	59.29	139.20	0.426	Collapse w/o circuit
Janus-Pro	decoupled	FAIL	FAIL ($A3$)	13.57	27.46	91.23	0.301	SigLIP cts. path

Table 15: Caption informativeness on COCO (VILA-U, 500 images). Complements the hallucination rates of Table 5: hallucination is reduced at a moderate informativeness cost, and METEOR improves.

Metric	Baseline	L_0 ablation
METEOR $\uparrow$	0.127	0.177
Object mentions / caption	6.02	5.38
Distinct objects / caption	2.36	1.88
Object recall (%) $\uparrow$	58.5	51.6